EFY TESTED S.C. Dwivedi

DUSK-TO-DAWN AUTO ON/OFF LED Strip Lighting With Timer

SURESH CHANDRA DWIVEDI

Automatic lighting systems are widely used to improve energy efficiency and convenience in residential and commercial applications. The presented auto dusk-to-dawn LED strip on/off system with timer circuit is designed to control LED strip lighting based on ambient light conditions. It ensures that the lights turn on at dusk (low light) and switch off at dawn (daylight), eliminating manual operation and reducing power consumption.

The circuit uses a light-dependent resistor (LDR) as the primary sensing element to detect changes in ambient light intensity. An operational amplifier functions as a comparator to process the LDR signal and generate a control output. This output drives a transistor stage, which in turn controls a timer IC configured as an astable multivibrator, providing stable switching action with optional delay. The timer function helps prevent flickering or false triggering caused by sudden light variations such as vehicle headlights or passing shadows.

The circuit also incorporates a transistor-based switching mechanism for efficient LED strip drive. It operates from a 12V DC supply, making it suitable for practical installations such as home lighting, garden illumination, signage, and decorative lighting systems. Overall, this system offers a simple, low-cost, and reliable solution for automated lighting with added timing control.

The circuit uses readily available components and can be assembled on a breadboard or a general-purpose PCB. Its simple, reliable operation, combined with adjustable timing settings, makes it well-suited for hobbyists, students, and practical lighting automation applications. Fig. 1 shows the prototype assembled on a breadboard.

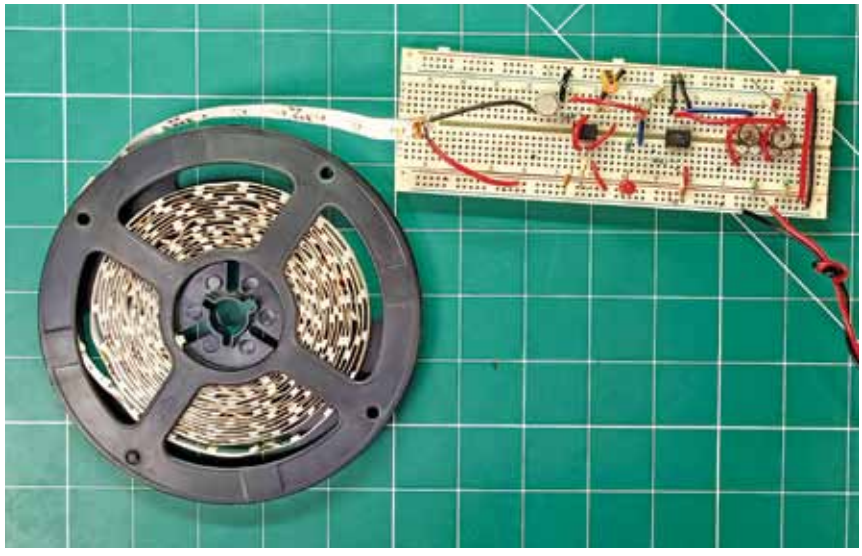


Fig. 1: Author's prototype

Circuit and working

Fig. 2 shows the circuit diagram of the auto dusk-to-dawn LED strip on/off system with a timer. It is built around an LDR (light-dependent resistor) to sense ambient light intensity, an LM311 comparator to detect the threshold level (preset voltage), and an NE555 timer to provide a selectable time delay. The LED strip driver transistor T2 is used to safely control a 12V LED strip.

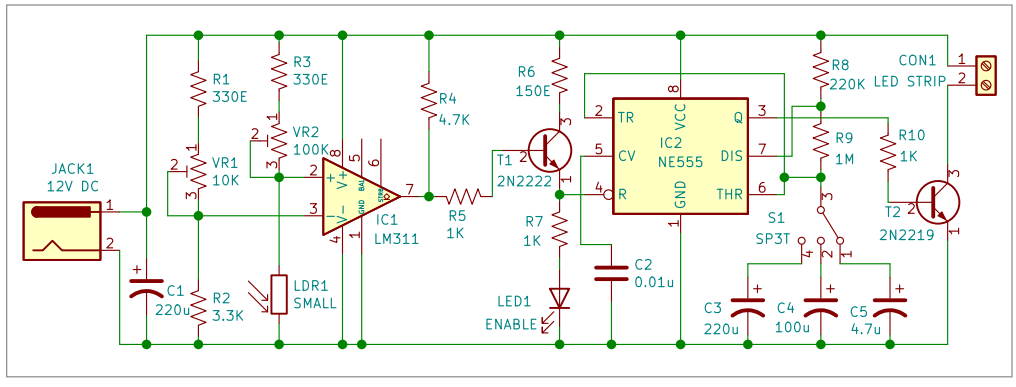


Fig. 2: Circuit diagram

A regulated 12V DC supply is applied through DC jack JACK1. Capacitor C1 filters